

Assignment - 3

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Page No.

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Short answer questions:

① What is the purpose of a machine learning model?

→ At its core, an ML model is a mathematical approximation of a real-world process. It learns patterns from historical data so it can make predictions on unseen data. The whole point is generalization, not just memorization.

A model is a function $f: x \rightarrow y$

Goal is to minimize a loss function $L(f, y)$.

Example: predicting housing prices using size, no of bedrooms, and location.

② Define stratified k-fold cross-validation with an example?

→ Standard k-fold CV splits data into k equal folds, but if the dataset is imbalanced, folds may not represent the whole dataset. Stratified ensures the same class proportions across data. Dataset divided into k folds. Each fold keeps label proportions consistent with the full dataset.

ex: In medical diagnosis (90% healthy, 10% sick), stratified or ensure each fold has the same 9:1 ratio, preventing biased performance estimation.

③ What is the time complexity of computing mean and variance?

→ Both computations are linear because they require looking at every data point. variance requires the mean first.

$$\text{Mean} = \mu = \frac{1}{n} \sum_{i=1}^n x_i \rightarrow O(n)$$

$$\text{variance} = \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 \rightarrow O(n)$$

ex: with 1M sample, both mean and variance scale linearly → fast enough for big data.

④ Explain the concept of one-hot encoding in feature engineering?

→ Converts categorical variables into binary vectors, preventing models from thinking categories have numeric order. If a categorical variable has k categories then it produces k binary features.

ex: car brand = {Toyota, BMW, Tesla} - "BMW" → [0, 1, 0],

original	one-hot encoded
Gender	Gender male female
male	male 1 0
female	female 0 1
male	male 1 0
female	female 1 0

⑤ What is the space complexity of a feature-engineered dataset?

→ Adding features increases dataset dimensionality. High dimensional data causes more memory / storage usage (curse of dimensionality).

$$\text{Storage} = O(n \times d)$$

Ex: one-hot encoding 1000 categories for 10,000 rows gives 10M values stored.

⑥ Define a test set with an example?

→ A separate dataset not touched during training or validation. It's the final exam for the model. Data split ex: Train 70%, Validation 15%, Test 15%.

Example:

In Netflix recommendations, test data = a hidden set of user viewing patterns - used only for final evaluation.

⑦ What is the purpose of data quality in machine learning?

→ A model is only as good as the data it learns from. Noisy, biased, or incomplete data makes the model unreliable. Noise inflates variance, bias reduces accuracy.

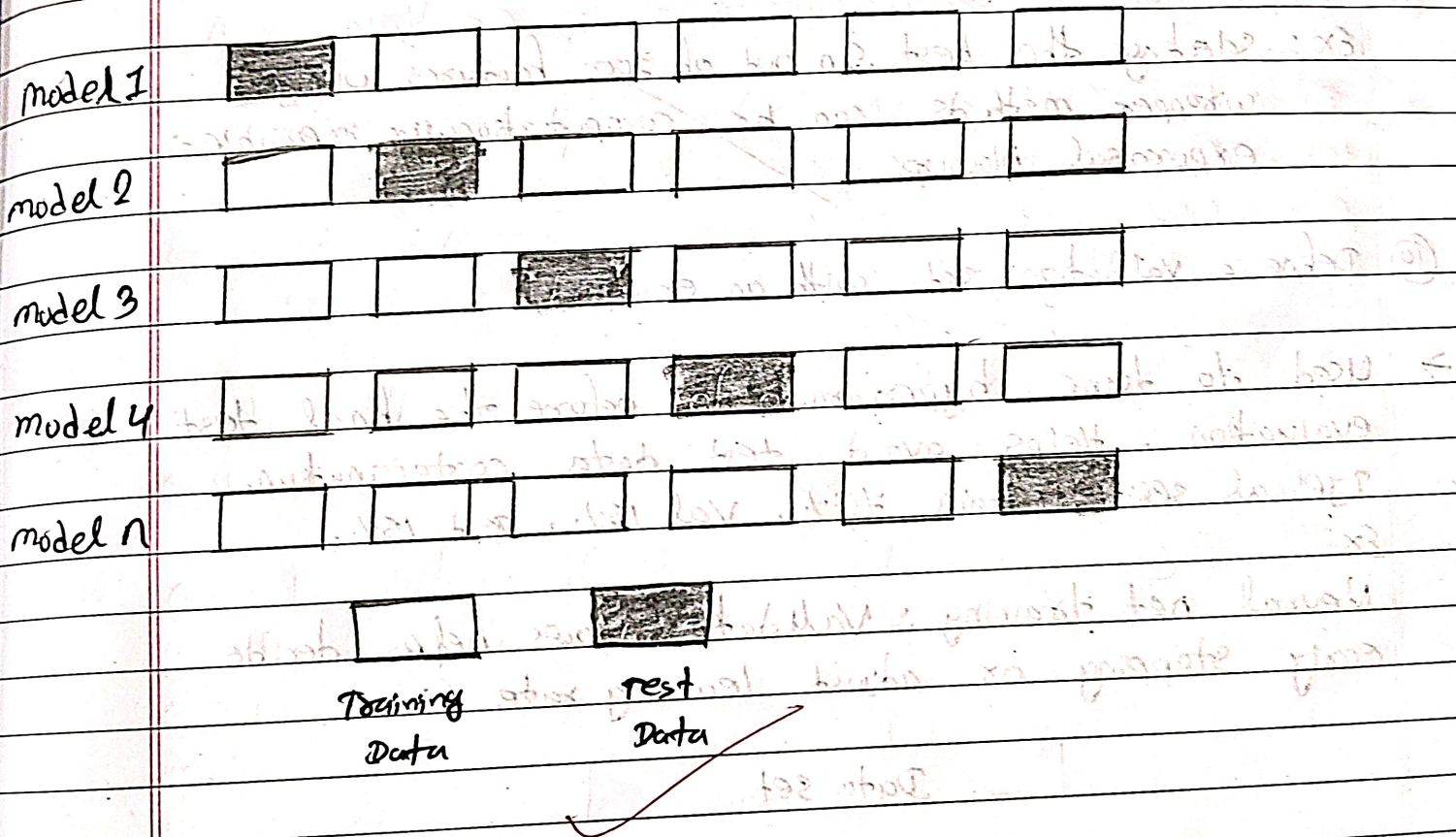
Ex: If a facial recognition dataset only contains lighter skin tones then there will be poor performance on darker skin: data quality failure.

② Explain the concept of leave-one-out cross-validation?

→ Extreme version of cv where each sample is used once as a test set. High accuracy, but very slow for large n . Here, $k=n$, complexity $\sim O(n^2)$.

Ex:

with 100 data point we train 100 models, each time leaving one sample out.



⑨ What is the time complexity of features selection?

→ Different methods:

- Filters: use statistics (correlation, chi-square).
- Wrapper: test subsets of features → slow.
- Embedded: selection inside training (e.g. Lasso).

Complexity: Filters: $O(nd)$

Wrapper: $O(2^d)$ worst-case

Embedded: $\sim O(nd)$

Ex: Selecting the best 20 out of 100 features using wrapper methods can be computationally impossible → exponential blowup.

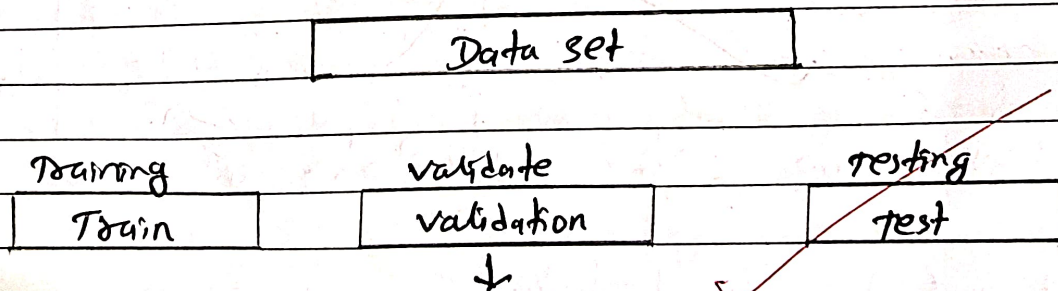
⑩ Define a validation set with an example?

→ Used to tune hyperparameters before the final test evaluation. Helps avoid test data contamination.

Typical split: Train 70%, Val 15%, Test 15%.

Ex:

Neural net training: Validation loss helps decide early stopping or adjust learning rate.



tune Hyper parameters and select Best model

⑪ Define the K-Nearest Neighbour algorithm?

→ Non-parametric, lazy learner. Predictions are made by comparing a new point to stored training points.

For classification: $\hat{y} = \text{model}(y_i \text{ of } k \text{ nearest neighbours})$

ex:

Predicting if a movie will be liked by a user based on k nearest users with similar preferences.

⑫ What is the time complexity of training a linear regression model?

→ Theory: Two training approaches:

- Normal Equation: closed-form, exact but expensive
- Gradient Descent: iterative, scales better.

Math:

- Normal equation: $O(d^3)$ due to matrix inversion
- Gradient Descent: $O(ndk)$, with k = iterations.

ex: for small d (features), Normal eqn is fine for text data (thousand of features), GD is better.

⑬ Explain the concept of a confusion matrix?

→ Summarize classification outcomes into true/false/positive/negative. used to compute accuracy, precision, recall and $F1$:

• Accuracy = $(TP + TN) / \text{Total}$

• Precision = $TP / (TP + FP)$

• Recall = $TP / (TP + FN)$

Q14) What is the purpose of the SSE metric in regression?

→ SSE measures how badly predictions differ from true labels. Models aim to minimize SSE during training.

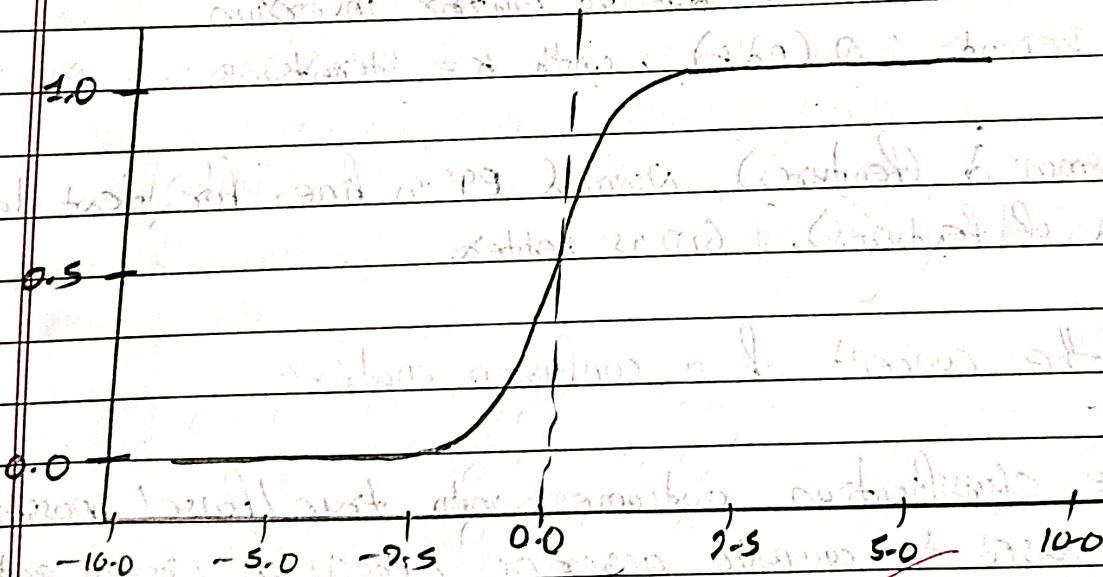
$$SSE = \sum (y_i - \hat{y}_i)^2$$

Ex: In predicting stock prices, smaller SSE means predictions are closer to reality.

Q15) Define logistic regression with an example?

→ Despite its name, it's a classification algorithm. Uses the sigmoid function to map values into probability.

$$\sigma(x) = \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-(w \cdot x + b)}}$$



Ex: predicting if a student passes (1) or fails (0) based on the study hours and attendance.

Long Questions:

① Discuss the role of cross-validation in preventing overfitting.

→ Cross validation is a model validation technique that splits data into training and validation sets multiple times to evaluate model generalization.

A single train-test-split may give misleading result. The model may overfit (memorize) training data, failing on unseen data.

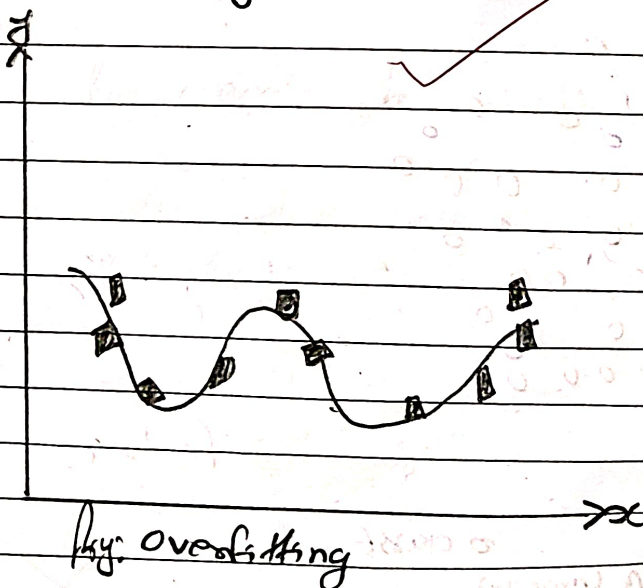
$$\text{Formula (k-fold cv error)} = \frac{1}{k} \sum_{i=1}^k \text{Error}_i$$

Types of CV:

- k-fold (most common)
- Leave-one-out (LOOCV)
- Stratified (for imbalanced datasets)

Ex: If you split data into 5 folds and the average accuracy is 92%, the model is likely generalizing well.

Conclusion: CV reduces overfitting by ensuring performance consistency across different subsets of data.



② Explain the implementation of k-nearest Neighbour for classification with an example?

→ KNN is a lazy, non-parametric algorithm that classifies a sample based on majority vote of its k nearest neighbours

Steps:

- 1) Choose k (odd number preferred)
- 2) Compute distance (Euclidean most common):

$$d(x_i, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

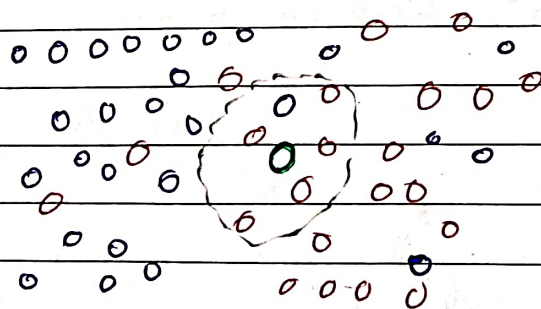
- 3) Select k nearest neighbours
- 4) Perform majority voting.

Ex: Classifying fruit based on weight & texture. If $k=3$.

• Distances → 2 apples, 1 orange → Predicted = Apple.

• Conclusion: KNN is simple, effective, but computationally expensive for large datasets.

Feature 2



Feature 1

○ class A

○ = ?

● class B

class B (majority)

③ Discuss the time and space complexity of linear regression with examples?

→ Two implementations:

1) Normal Equation (closed form):

• Formula:

$$\theta = (X^T X)^{-1} X^T y$$

• Time: $O(n^2 m + n^3)$

• Space: $O(n^2)$

• works well for small n .

2) Gradient Descent:

• update rule:

$$\theta := \theta - \alpha \frac{1}{n} X^T (X\theta - y)$$

• Time: $O(mnk)$, where k = iterations.

• Space: $O(n)$

Example:

For 10,000 samples & 50 features → Gradient descent is faster than inverting a 50×50 matrix.

Normal equation is good for small features sets, while gradient descent is scalable.

② Describe the role of evaluation metrics in supervised learning with examples?

→ Accuracy can be misleading (e.g. 95% accuracy but misses all fraud cases if only 5% fraud exists).

• Metrics & Formulas:

$$\text{Accuracy} = \frac{TP + TN}{(TP + TN + FP + FN)}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

• ROC-AUC = Area under True Positive Rate vs False Positive Rate curve.

Example: In medical diagnosis:

- Recall is crucial (catch all sick patients).
- Precision: avoid false alarms.

• Conclusion: Choice of metric depends on application.

⑤ Explain the implementation of a ~~confusion matrix~~ logistic regression for binary classification?

→ Logistic Regression models probability of output belonging to class 1.

Formula Sigmoid:

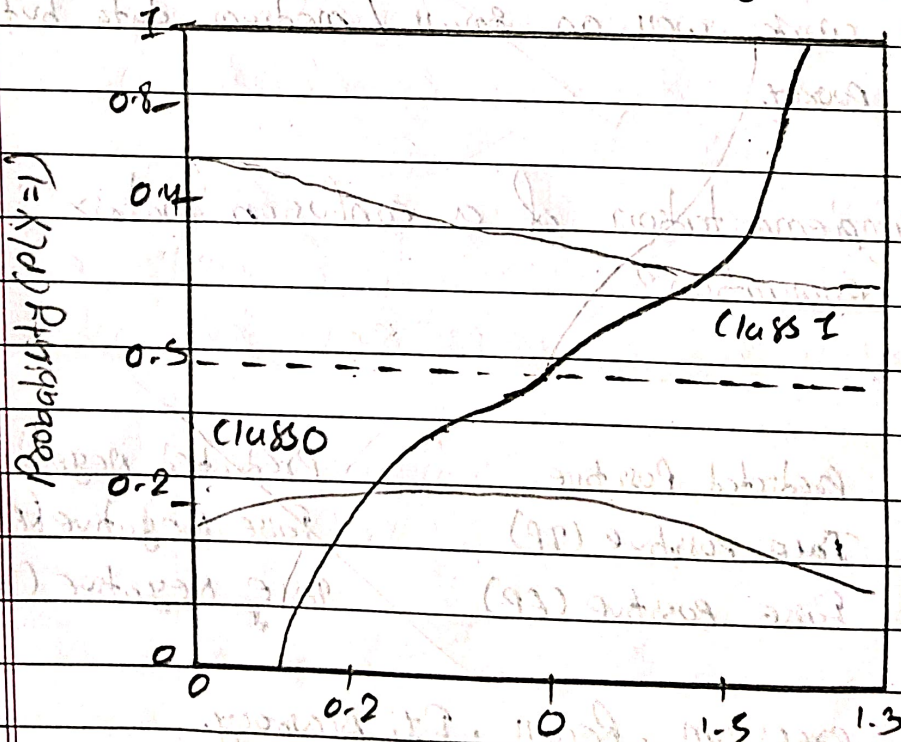
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$= \frac{1}{1 + e^{-(w \cdot x + b)}}$$

- Decision: if $\sigma > 0.5 \rightarrow \text{class 1}$ else class 0.
- Training: maximize likelihood using gradient descent.
- Example: Predicting if a student passes based on hours studied:

• $P = 0.82 \rightarrow$ student predicted to "Pass".

- Conclusion: Logistic regression is simple, interpretable, and widely used for binary classification.



Key: Linear combination of inputs (z)

⑥ Discuss the time and space complexity of Support Vector machines?

→ Training:

- Linear SVM: $O(mn)$ to $O(m^2n)$.
- Kernel SVM: $O(m^2n)$ to $O(m^3)$ (due to quadratic optimization)

Prediction:

- $O(s \cdot n)$, where s = support vectors.
- space: $O(m^2)$ storing kernel matrix
- Examples: for 50000 samples, training with kernel SVM is impractical, so linear / approximate methods are used
- Conclusion: SVMs work well on small / medium data but scale poorly.

⑦ Describe the implementation of a confusion matrix for classification evaluation?

→ structure:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- Derived metrics: precision, Recall, F1, Accuracy.

- Example (Disease prediction):
 - TP = 50 (correctly predicted sick)
 - FP = 10 (healthy marked sick)
 - FN = 5 (sick missed)
 - TN = 100 (healthy predicted correctly)
- Conclusion: confusion matrix gives detailed error analysis beyond accuracy.

⑧ Explain the importance of precision, recall and F-score in classification?

-
- Precision: out of predicted positive, how many are correct.
 - Recall (Sensitivity): out of actual positives, how many are detected.
 - F1-score: Balance between precision & recall.

Formulas:

$$\text{Precision} = \frac{TP}{TP + FP}$$

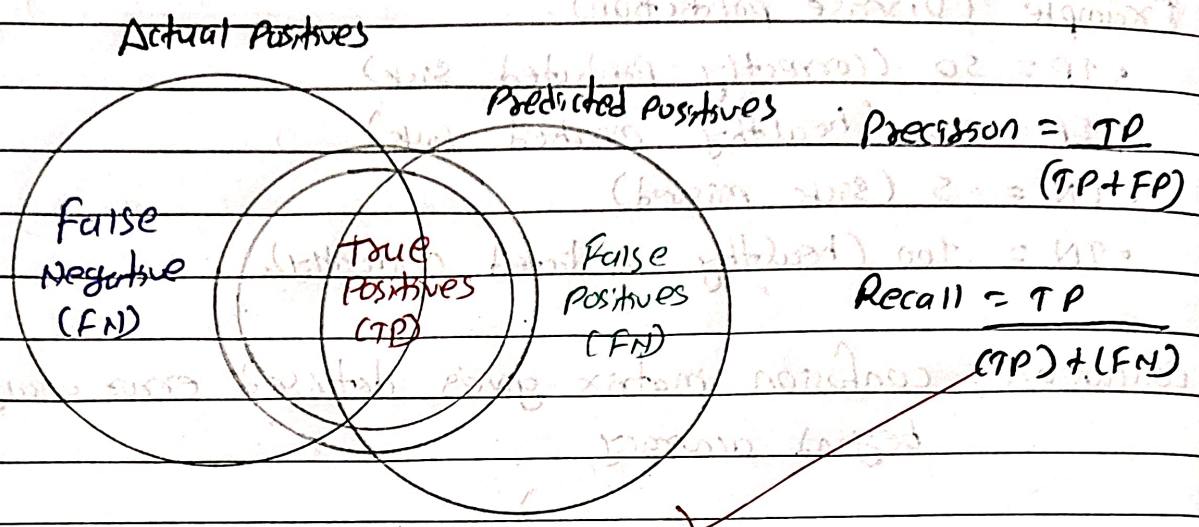
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2PR}{P+R}$$

Example:

- Fraud detection → Recall important.
- Medical test → Precision important.
- Imbalanced dataset → F1-score is best.

Conclusion: Different applications prioritize different metrics.



Q) Discuss the applications of supervised learning in real-world scenarios?

→ Application are as:-

- i) Health care: Disease prediction (logistic regression).
- ii) Finance: credit scoring, fraud detection (svm, decision tree)
- iii) Retail: Product recommendation, demand prediction
- iv) Autonomous cars: Traffic sign recognition (CNNs)
- v) Example: Email spam filtering using Naive Bayes classifier

Supervised learning is used in nearly all industries where labeled data is available.

⑩ Discuss the role of cross-validation in preventing overfitting?

⇒ Cross-validation (CV) is a model validation technique that splits data into training and validation sets multiple times to evaluate model generalization.

A single train-test split may give misleading results. The model may overfit (memorize) training data, failing on unseen data.

$$k\text{-fold CV error} = \frac{1}{k} \sum_{i=1}^k \text{Error}_i$$

Types:

- k-fold CV
- Stratified k-fold (for imbalanced data)
- leave-one-out CV

Example: Fraud detection → Stratified CV ensures equal ratio of fraud / non-fraud in each fold.

Cross validation ensures stability, prevents overfitting and provides reliable model.

[Signature]